DSC630-T302 | PREDICTIVE ANALYTICS

Predicting Customer Churn in Telecommunications

A machine learning approach to identifying at-risk customers before they cancel

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THE PROBLEM

Retention Is Cheaper Than Acquisition

Can customer demographics, service subscriptions, and billing data predict which customers are about to cancel, and what should the company do about it?

CENTRAL BUSINESS QUESTION

5–7x more costly to acquire a new customer than to retain one (Siddiqi, 2012)

26.5% of customers in this dataset churned, a real and recurring pattern

WHY IT MATTERS TO EACH STAKEHOLDER



Marketing & Retention Teams



Product Managers



Customer Service Leadership



Executive Leadership

THE DATA

7,043 Customers, 21 Features, One Outcome



7,043

customer records

21

columns per customer

Source: IBM Telco Customer Churn dataset (IBM Watson Analytics / Kaggle mirror)



Demographics

4 features

Gender, senior citizen status, partner, dependents



Service Subscriptions

9 features

Phone, internet type, security/backup/support add-ons, streaming



Account Information

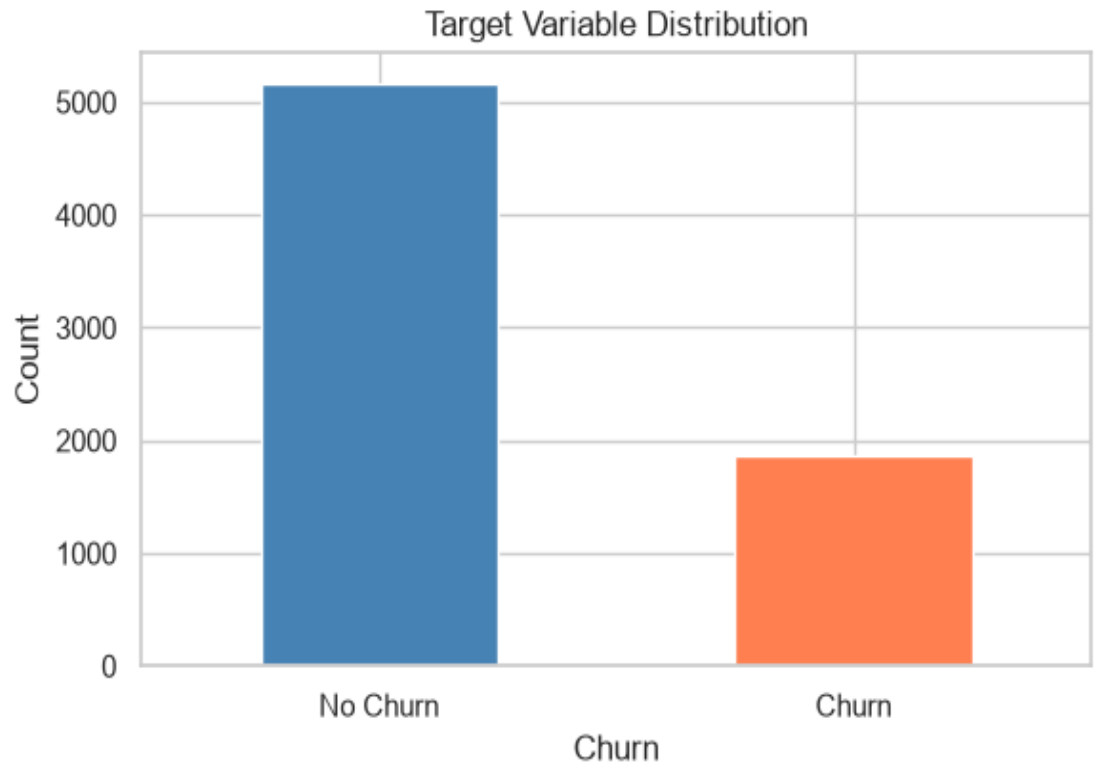
4 features

Tenure, contract type, paperless billing, payment method, charges

Data quality: 11 blank TotalCharges values, all tenure = 0 (brand-new customers), imputed using MonthlyCharges rather than dropped.

EXPLORATORY ANALYSIS

A Real, but Manageable, Imbalance



73.5% of customers stayed with the company

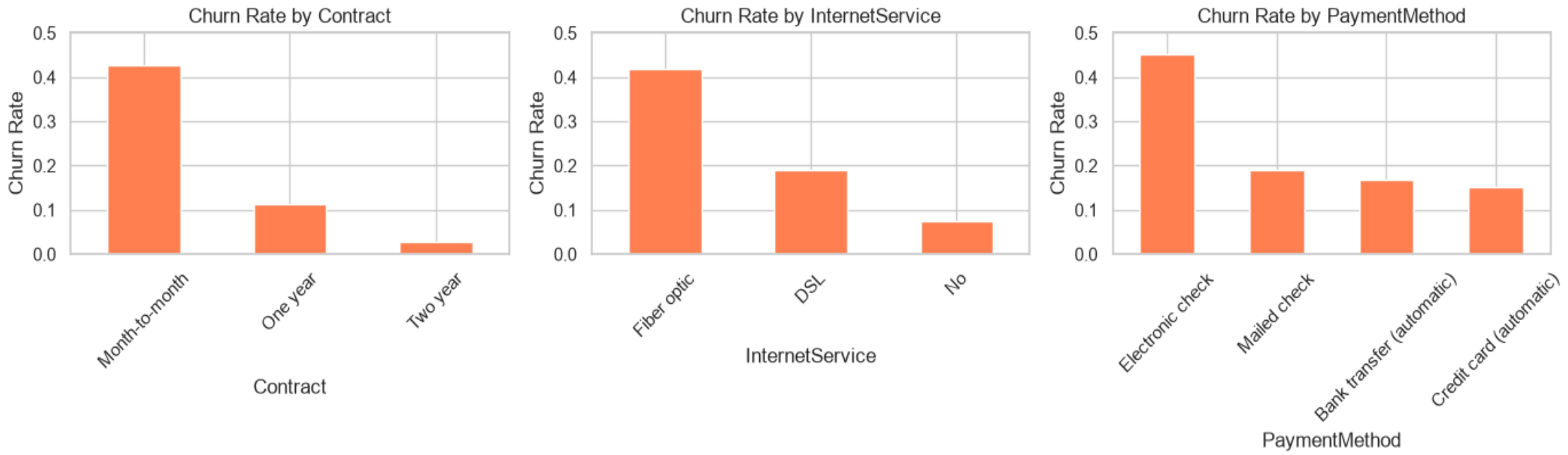
26.5% of customers churned, not a rare event, and worth modeling carefully

This imbalance shaped the modeling strategy: a stratified train/test split kept the same churn ratio in both sets, and balanced class weights were used instead of oversampling.

Sets up why raw accuracy is a misleading metric here. Revisited in the Results section.

EXPLORATORY ANALYSIS

Contract Type Is the Clearest Signal

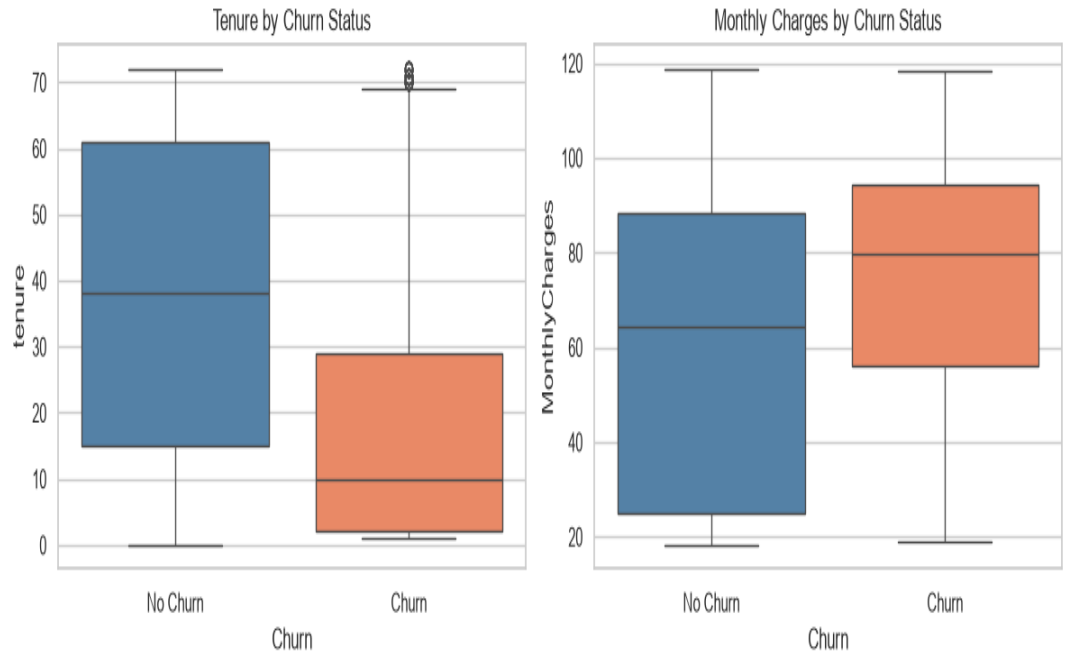


Month-to-month customers churn at 43% versus 11% for one-year contracts and just 3% for two-year contracts.

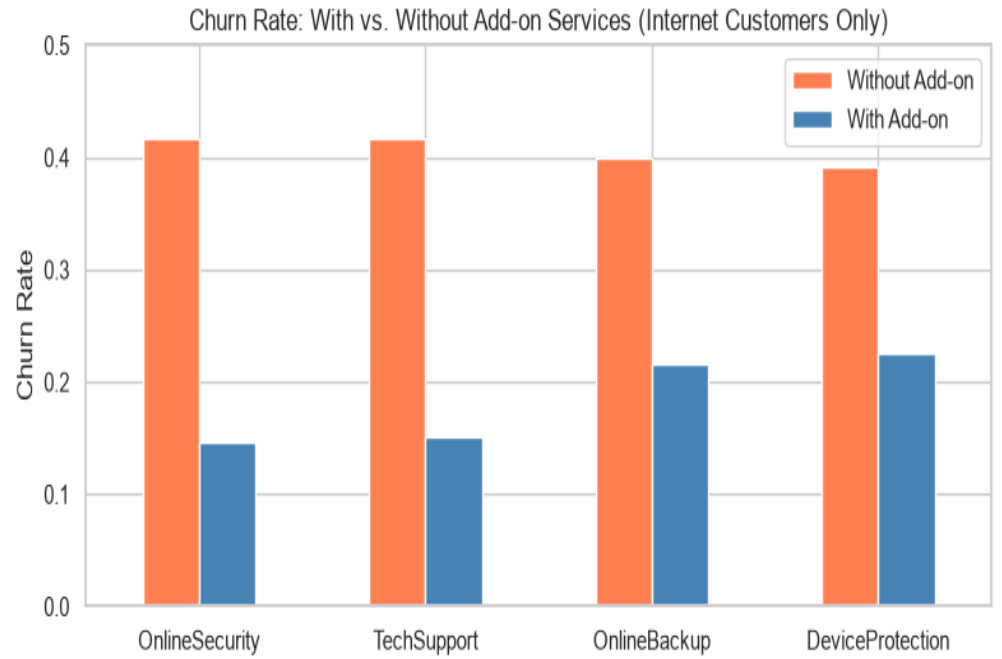
EXPLORATORY ANALYSIS

New Customers and Unprotected Accounts Are Highest Risk

Tenure & Monthly Charges by Churn Status



Churn Rate: With vs. Without Add-on Services



Churned customers have a median tenure of about 10 months vs. 38 for those who stay, and add-ons like security and support roughly halve churn risk.

METHODOLOGY

Four Models, One Fair Comparison

PREPROCESSING PIPELINE

1. Drop customer ID (no predictive value)
2. Clean & impute TotalCharges
3. One-hot encode categoricals → 30 binary features from 16 columns
4. 80/20 stratified split (5,634 train / 1,409 test)
5. Scale tenure, MonthlyCharges, TotalCharges
6. Balanced class weights instead of resampling

MODELS TRAINED

Naive Baseline	Establish a performance floor
Logistic Regression	Interpretable coefficients for business insight
Decision Tree	Non-linear relationships, visualizable rules
Random Forest	Ensemble accuracy, captures feature interactions

SUCCESS CRITERIA (TEST SET)

- Recall ≥ 0.70
- AUC-ROC ≥ 0.80
- Precision ≥ 0.50
- F1 ≥ 0.60

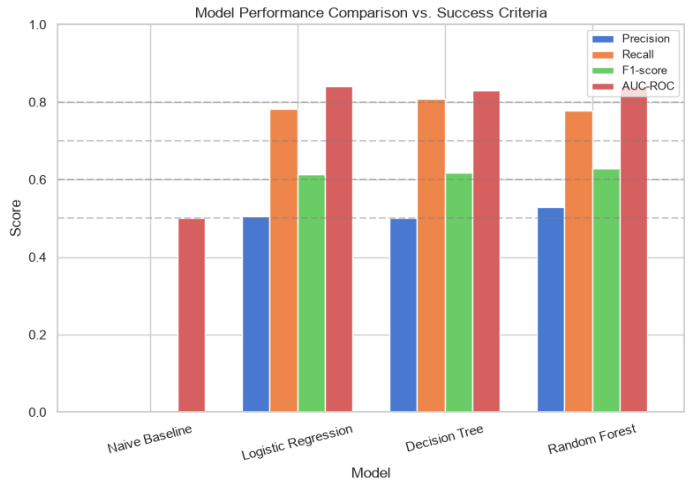
RESULTS

Every Model Cleared the Bar

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Naive Baseline	0.7346	0.0000	0.0000	0.0000	0.5000
Logistic Regression	0.7381	0.5043	0.7834	0.6136	0.8418
Decision Tree	0.7346	0.5000	0.8075	0.6176	0.8308
Random Forest	**0.7566**	**0.5281**	**0.7781**	**0.6292**	**0.8409**

Random Forest delivered the best overall balance of precision, recall, and F1-score.

The naive baseline reaches 73.5% accuracy by always guessing “no churn,” but with zero recall it never catches a single churner, which is why accuracy alone is the wrong metric for this problem.



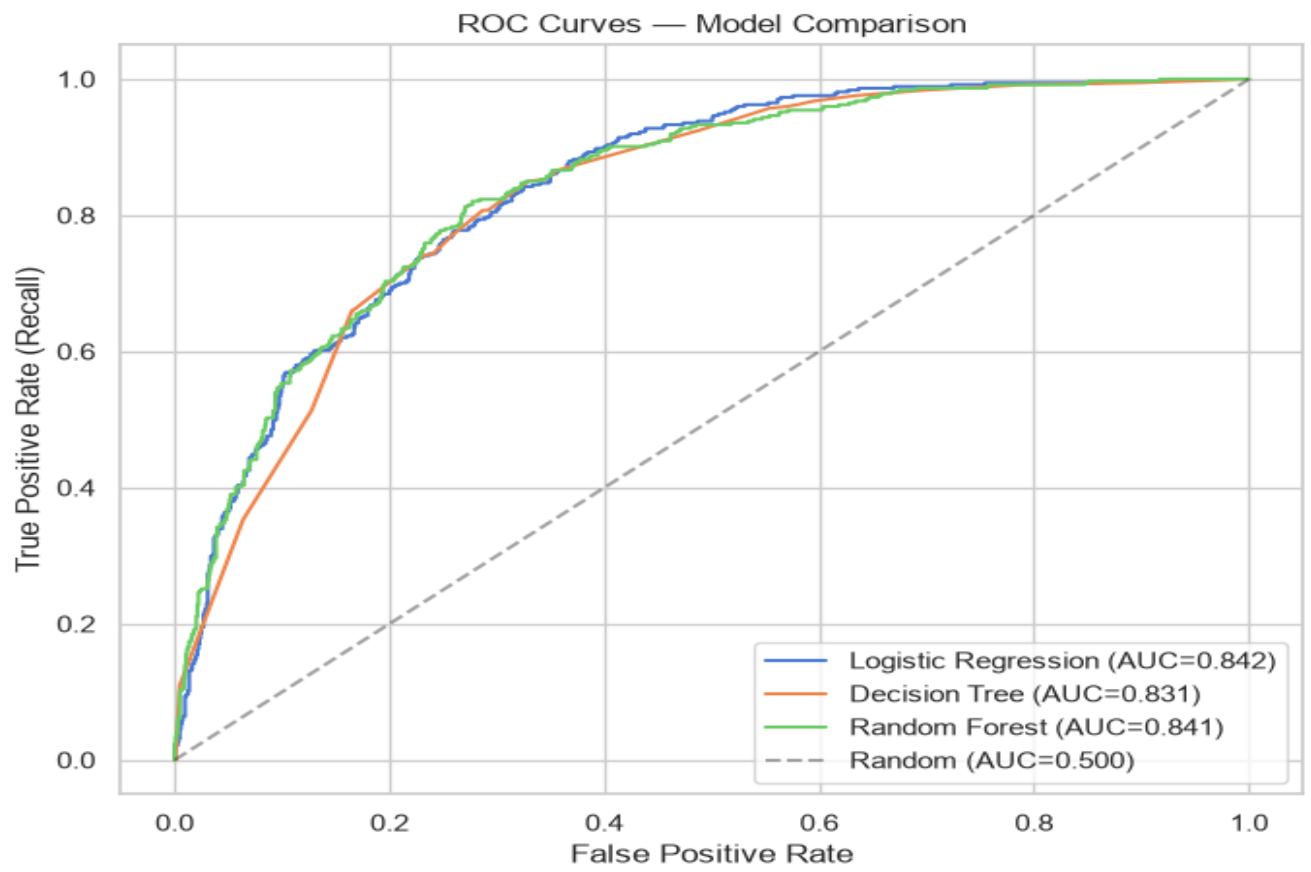
Recall, precision, F1, and AUC-ROC vs. success thresholds

291 of 374

test-set churners correctly flagged by Random Forest (78% recall)

RESULTS

All Three Models Separate Churners Well



0.842 Logistic Regression AUC-ROC

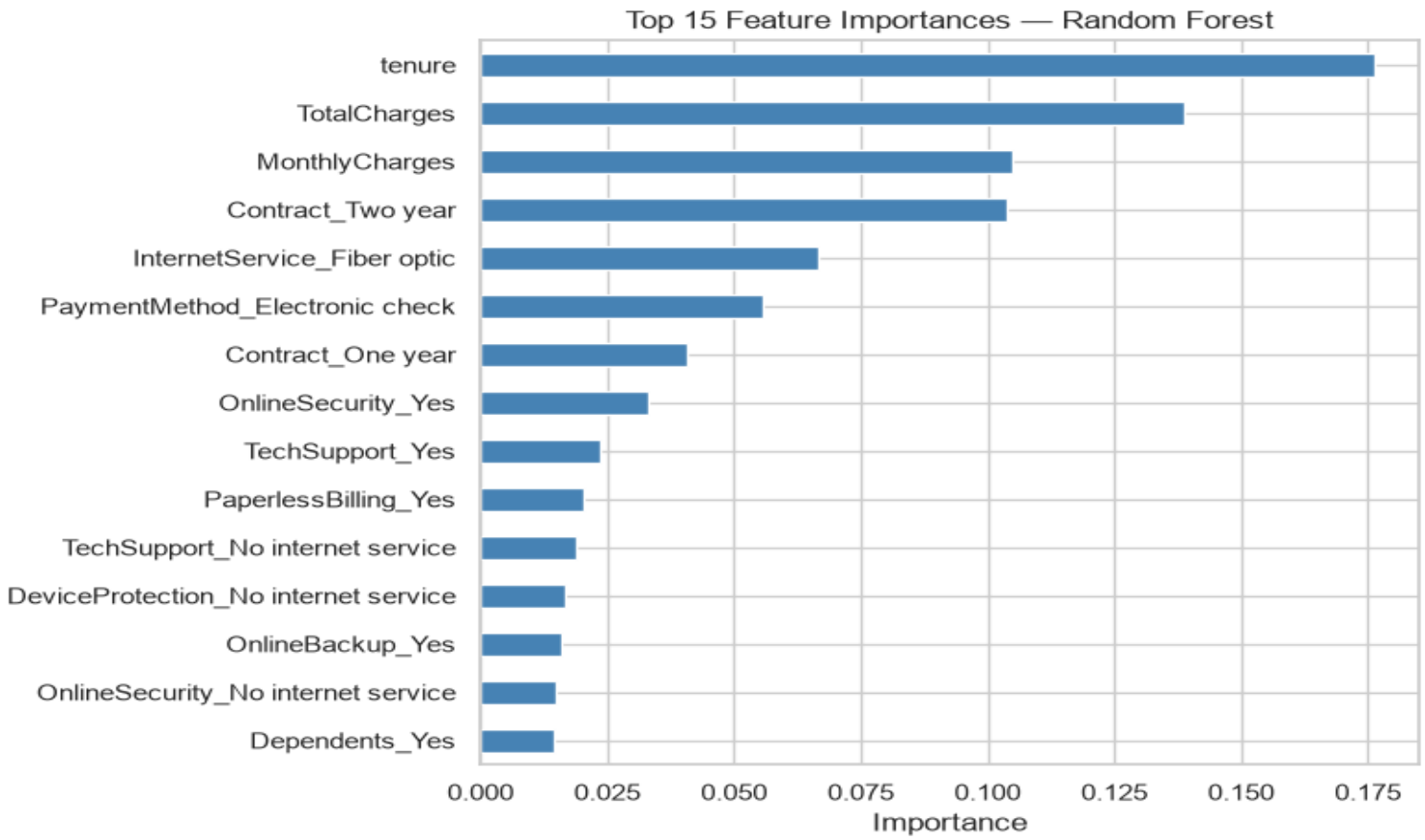
0.841 Random Forest AUC-ROC

0.831 Decision Tree AUC-ROC

A score near 0.50 (the dashed line) means random guessing. All three models sit well above 0.80.

RESULTS

Tenure, Charges, and Contract Type Drive the Model



WHAT THE MODEL LEANS ON

Random Forest feature importances show which inputs most influence each prediction.

1. **Tenure**
2. **Total Charges**
3. **Monthly Charges**

Every one of these top features maps to something a retention team can see on a customer's account, and potentially act on.

KEY FINDINGS

Six Drivers Explain Most of the Churn Risk



Contract type

Month-to-month churns ~43% vs. ~3% for two-year contracts



Tenure

The first 12 months is the highest-risk window



Internet service

Fiber optic customers churn more than DSL customers



Add-on services

No security, support, or backup roughly doubles churn risk



Payment method

Electronic check users churn more than automatic-pay customers



Monthly charges

Higher bills push churn up, especially with no contract

All six drivers are visible in data the company already collects, which is what makes each one actionable rather than theoretical.

RECOMMENDATIONS

Five Targeted Actions, Prioritized by Impact

HIGH	**Offer contract upgrade incentives**	*Month-to-month customers*	Contract type is the #1 churn driver
HIGH	**Proactive outreach in first 12 months**	*New customers (tenure < 12 mo)*	Early tenure is the highest-risk window
MED	**Bundle add-on services at a discount**	*Customers without security/support*	Add-ons strongly associated with retention
MED	**Strengthen competitive positioning**	*Fiber optic subscribers*	Higher churn driven by market competition
LOW	**Encourage automatic payment enrollment**	*Electronic check users*	Manual payment associated with higher churn

ETHICAL CONSIDERATIONS

Predictive Power Comes With Responsibility



Demographic fairness

Gender showed no meaningful disparity; senior citizens showed a modest recall gap worth monitoring



Data privacy

Billing/service data was collected for service delivery, so using it for retention targeting should be disclosed



Intervention ethics

Retention offers must be applied equitably, not skewed by demographic group



Transparency

At-risk customers should get a better offer, not manipulative pressure tactics



Model drift

Customer behavior changes over time; a stale model can develop new biases

MITIGATIONS

Quarterly fairness audits • Clear privacy disclosures • Human oversight on high-impact decisions • Scheduled model retraining

CONCLUSION

Churn Is Predictable, and Now It's Actionable

0.84

AUC-ROC (Random Forest)

0.78

Recall (Random Forest)

0.63

F1-Score (Random Forest)

Churn is concentrated in identifiable, influenceable segments: contract type, tenure, service type, and add-ons, all of which the business can act on directly.

NEXT STEPS



Pilot deployment with threshold tuning for campaign cost



A/B test model-flagged customers vs. random outreach



Add gradient boosting and engineered interaction features



Periodic retraining with drift monitoring